Countable and Uncountable Sets Solutions

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Exercises

1. Show that $\mathbb{Q}$ is countably infinite.

Proof. Recall $\mathbb{Z}$ is countably infinite. Note $\mathbb{Z}-\{0\}$ is an infinite subset of $\mathbb{Z}$, and thus by Theorem 7.3 the set $\mathbb{Z}-\{0\}$ is countable. Furthermore, the set $\mathbb{Z}\times(\mathbb{Z}-\{0\})$ is also countable by Theorem 7.6. Thus, define $f : \mathbb{Z}\times(\mathbb{Z}-\{0\}) \to \mathbb{Q}$ by the equation $f(n,m) = n/m$. Note that f is a surjection since for any $a/b \in \mathbb{Q}$, $a \in \mathbb{Z}$ and $b \in \mathbb{Z}-\{0\}$. Given that $\mathbb{Z}\times(\mathbb{Z}-\{0\})$ is countable, there exists a surjection $g : \mathbb{Z}_+ \to \mathbb{Z}\times(\mathbb{Z}-\{0\})$. As such, $f \circ g : \mathbb{Z}_+ \to \mathbb{Q}$ is a surjection. Therefore, $\mathbb{Q}$ is countable. Since $\mathbb{Z} \subset \mathbb{Q}$, we further conclude $\mathbb{Q}$ is countably infinite. □

2. Show that the maps f and g of Examples 1 and 2 are bijections.

Proof. The map $f : \mathbb{Z} \to \mathbb{Z}_+$ in Example 1 is defined as

$$f(n) = \begin{cases} 2n & n > 0, \\ -2n+1 & n \leq 0. \end{cases}$$

To show f is an injective map, let $m, n \in \mathbb{Z}$ such that $f(n) = f(m) = a$. In particular, note m and n must either be both positive or both non-positive. This is because if without loss of generality n is positive and m is non-positive we have $f(n) = 2n$ and $f(m) = -2m+1$. In particular, $2n = -2m+1 = a$, which cannot be true since $2n$ is even and $-2m+1$ is odd, and a positive integer cannot be both even and odd. Thus, if both n and m are positive, then $2n = 2m$ by assumption, which implies $n = m$. Hence, f is injective. If n, m are both non-positive, then $-2n+1 = -2m+1$. This implies $-2n = -2m$, and thus $n = m$, which allows us to conclude f is injective.

To show f is surjective, consider $a \in \mathbb{Z}_+$. If a is even, $a = 2n$ for some $n \in \mathbb{Z}_+$, as such there exists such $n \in \mathbb{Z}$ that is positive and is mapped by f onto a. If a is odd, let $n = (1-a)/2$. Since a is odd, $1-a$ is equivalent to $1-2k-1$ for some $k \in \mathbb{Z}$, and thus $1-a$ is an even integer, implying $(1-a)/2$ is also an integer. Since $a \geq 1$, $1-a \leq 0$, implying $(1-a)/2 \leq 0$ also. Thus, $n \in \mathbb{Z}$ is a non-positive integer which is mapped by f onto a. Therefore, f is surjective, and thus we conclude f is a bijection.

The map $g : A \to \mathbb{Z}_+$ is defined by the formula

$$g(n,m) = \frac{1}{2}(n-m)n + m,$$

where A is the subset of $\mathbb{Z}_+\times\mathbb{Z}_+$ consisting of pairs (n,m) for which $m \leq n$. ⋆**Haven't proved that this g function is a bijection.**⋆ □

3. Let X ve the two-element set $\{0, 1\}$. Show there is a bijective correspondence between the set $\mathcal{P}(\mathbb{Z}_+)$ abd the cartesian product X^ω.

4.

(a) A real number x is said to be **algebraic** (over the rationals) if it satisfies some polynomial equation of positive degree

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$$

with rational coefficients a_i. Assuming that each polynomial equation has only finitely many roots, show that the set of algebraic numbers is countable.

(b) A real number is said to be **transcendental** if it is not algebraic. Assuming that the reals are uncountable, show that the transcendental numbers are uncountable. (It is a somewhat surprising fact that only two transcendental numbers are familiar to us: e and π. Even proving these two numbers transcendental is highly non-trivial).

5. Determine, for each of the following sets, whether or not it is countable. Justify your answers.

(a) The set A of all functions $f : \{0, 1\} \to \mathbb{Z}_+$.
(b) The set B_n of all functions $f : \{1, \ldots, n\} \to \mathbb{Z}_+$.
(c) The set $C = \bigcup_{n \in \mathbb{Z}_+} B_n$.
(d) The set D of all functions $f : \mathbb{Z}_+ \to \mathbb{Z}_+$.
(e) The set E of all functions $f : \mathbb{Z}_+ \to \{0, 1\}$.
(f) The set F of all functions $f : \mathbb{Z}_+ \to \{0, 1\}$ that are "eventually zero." [We say that f is **eventually zero** uf there is a positive integer N such that $f(n) = 0$ for all $n \geq N$].
(g) The set G of all functions $f : \mathbb{Z}_+ \to \mathbb{Z}_+$ that are eventually 1.
(h) The set H of all functions $f : \mathbb{Z}_+ \to \mathbb{Z}_+$ that are eventually constant.
(i) The set I of all two-element subsets of $\mathbb{Z}_+$.
(j) The set J of all finite subsets of $\mathbb{Z}_+$.

6. We say that two sets A and B **have the same cardinality** if there is a bijection of A with B.

(a) Show that if $B \subseteq A$ and if there is an injection

$$f : A \hookrightarrow B,$$

then A and B have the same cardinality. [Hint: Define $A_1 = A$, $B_1 = B$, and for $n > 1$, $A_n = f(A_{n-1})$ and $B_n = f(B_{n-1})$. Note that $A_1 \supseteq B_1 \supseteq A_2 \supseteq B_2 \supseteq \cdots$. Define a bijection $h : A \to B$ by the rule

$$h(x) = \begin{cases} f(x) & \text{if } x \in A_n - B_n \text{ for some } n, \\ x & \text{otherwise.} \end{cases}]$$

(b)

Theorem. *(Schroeder-Bernstein Theorem) If there are injections $f : A \hookrightarrow C$ and $g : C \hookrightarrow A$, then A and C have the same cardinality.*

7. Show that the sets D and E of Exercise 5. have the same cardinality.

8. Let X denote the two-element set $\{0, 1\}$; let $\mathcal{B}$ be the set of countable subsets of X^ω. Show that X^ω and $\mathcal{B}$ have the same cardinality.

9.

(a) The formula

$$\begin{aligned} h(1) &= 1 \\ h(2) &= 2 \\ h(n) &= [h(n+1)]^2 - [h(n-1)]^2 \qquad n \geq 2 \end{aligned}$$

is not one to which the principle of recursive definition applies. Show that nevertheless there does exists a function $h : \mathbb{Z}_+ \to \mathbb{R}$ satisfying this formula. [Hint: Reformulate the formula so that the principle will apply and require h to be positive].

(b) Show that the formula in part (a) does not determine h uniquely. [Hint: If h is a positive function satisfying the formula let $f(i) = h(i)$ for $i \neq 3$, and let $f(3) = -h(3)$].

(c) Show that there is no function $f : \mathbb{Z}_+ \to \mathbb{R}$ satisfying the formula

$$\begin{aligned} h(1) &= 1 \\ h(2) &= 2 \\ h(n) &= [h(n+1)]^2 + [h(n-1)]^2 \qquad n \geq 2 \end{aligned}$$